Links in Linux – Hard Link & Soft Link

Complete Notes with Inode Examples

1. Hard Link

Create a Hard Link and check inode numbers:

```
echo "Hello Linux" > original.txt
ln original.txt hardlink.txt
ls -i
ls -li
```

2. Soft Link (Symbolic Link)

Create a Soft Link and check inode numbers:

```
echo "Linux World" > original.txt
ln -s original.txt softlink.txt
ls -i
ls -li
readlink softlink.txt
```

3. Difference Between Hard Link and Soft Link

Feature	Hard Link	Soft Link (Symlink)
Inode Number	Same as original	Different (points to path)
Works if file deleted?	Yes	No, broken
Across filesystems	Not allowed	Allowed
Points to	Actual data (inode)	File path
Command to verify	ls -i, ls -li	ls -li, readlink

4. Useful Commands

```
ls -i
ls -li
readlink softlink.txt
rm hardlink.txt
rm softlink.txt
```

Summary

- Use `ln file linkname` → Hard link
- Use `ln -s file linkname` → Soft link
- Use `ls -i` / `ls -li` to verify inode numbers
- Use `readlink` for symlink target